

Brian Wonjun Lee

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EDUCATION

University of Pennsylvania

MSE in Computer Graphics & Game Technology (CGGT), Submatriculation

BA in Computer Science & Design (Double Major), Benjamin Franklin Scholars

Relevant Coursework: Interactive Computer Graphics, Computer Animation, 3-D Computer Modeling, Introduction to Algorithms, Introduction to Computer Systems, Programming Languages & Techniques

Philadelphia, PA

Expected May 2028

Expected May 2027

EXPERIENCE

Software Engineer Intern, Aleph Lab (Y Combinator)

June 2026 – Aug 2026 · San Francisco (Remote)

- Delivered an app-wide **React Native** redesign from Figma to production, sequencing information architecture ahead of the visual system so users kept their bearings, then carrying a **33-section** design system onto every screen
- Automated a push-notification retention system that had required manual approval for every send**, leaving it inactive for weeks at a time — it now selects recipients, clears safety guardrails and delivers unattended at **91% device reach**
- Corrected eligibility logic that had been excluding most of the intended audience**, expanding reach **~2.7×** and producing the first paid conversion the re-engagement campaign had generated
- Made a children's app safe from notification injection** — anyone with a user ID could push arbitrary content until I bearer-authed four public endpoints, **zero caller changes**

Software Engineer Intern, BitMango (Puzzle1 Studio)

June 2025 – Aug 2025 · Pangyo, South Korea

- Maintained a live portfolio of **50+ Unity mobile game titles**, improving UI/UX and core gameplay stability across releases
- Resolved **100+ QA-reported bugs** by modifying Unity prefabs and C# scripts, shipping fixes in production releases
- Upgraded SDKs and platform modules fleet-wide to meet app-store requirements and sustain availability

Project Lead & Product Designer, Penn Spark

Jan 2025 – Present · Philadelphia, PA

- Led a cross-functional team of **15+** designers and developers to ship client-facing **0-to-1 products**
- Defined end-to-end user flows, interaction patterns, and reusable interface systems across multiple deliverables
- Shipped polished, user-centered features across semester-long client project cycles

Military Interpreter, Capital Defense Command (ROK Army)

Dec 2022 – June 2024 · Seoul, South Korea

- Provided real-time English–Korean interpretation in high-security, multi-agency environments for international stakeholders
- Translated sensitive documents and briefings with precision, enabling coordination across government and defense teams

Database Engineer Intern, IT-Farm

June 2022 – Aug 2022 · Seongnam, South Korea

- Designed and implemented a SQL database prototype managing **1M+ data entities** using Oracle and SQLite
- Processed and organized semiconductor datasets from **20+ client companies**, including Samsung, SK, and LG
- Wrote and optimized SQL queries to improve retrieval and management efficiency across large-scale datasets

PROJECTS

Mini Minecraft | C++, OpenGL, GLSL

Apr 2026

- Made a from-scratch voxel world read as varied rather than noisy — 7-biome procedural generation** (Perlin/FBM/Voronoi) and **3D Perlin caves** in a C++/OpenGL engine (team of 3)
- Made that world legible** — the GLSL pipeline I owned: PCF shadows, screen-space reflections, vertex AO, day-night cycle

Passenger | Unreal Engine 5, HLSL, Maya

Mar 2026 – Apr 2026

- Gave a short film its entire visual identity — 2 HLSL post-process shaders** (anisotropic Kuwahara, cell shader) over GBuffer channels, delivered as a ray-traced 2560×1080 render
- Brought its character to life** — rigged and animated in Maya (IK limbs, spline-IK spine) driving an Unreal Animation Blueprint

Mini Maya | C++, OpenGL, Qt

Feb 2026 – Mar 2026

- Made arbitrary meshes editable in place** — a **half-edge structure** giving topology-aware traversal, plus an OBJ parser accepting arbitrary n-gons, rendered via OpenGL VBOs
- Turned coarse geometry into smooth surfaces with Catmull–Clark subdivision** and topology ops in a Qt editor

TECHNICAL SKILLS

Graphics & Rendering: OpenGL, GLSL, HLSL, shader programming, Unreal Engine 5, Lumen, real-time & offline rendering

3D & Tooling: Maya, Blender, Substance Painter, Qt, Movie Render Pipeline

Languages: C++, Python, TypeScript, JavaScript, Java, Kotlin, C#, GLSL/HLSL, SQL

Frameworks, Infra & AI: React Three Fiber, Three.js, React Native (Expo), Node.js, Git, Docker, GCP, Claude Code, Codex, MCP

Practices: Trunk-based development, feature flags & staged rollout, CI/CD, code review, unit & regression testing